

DATA VISUALISATIONS

a simple guide



Navigating a Data-Driven World



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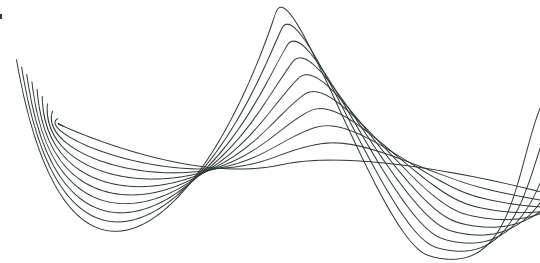


Introduction



We live in a world shaped by data. From the recommendations on your streaming service, to the graphs in your lecture slides, to the fitness stats on your phone - data is embedded in nearly every part of modern life. Yet for many university students, it can feel intimidating or abstract. Data literacy, the ability to read, interpret and communicate data, is one of the most transferable skills you can develop at university, and one that employers across every industry are actively looking for. This handbook offers a practical starting point.

Inside, you will find:



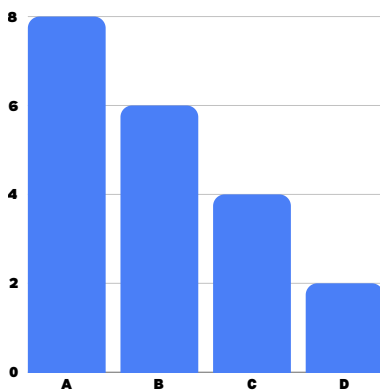
- **Data Visuals** - an introduction to the most common chart types you will encounter in research, media and everyday life
- **Real Applications** - where data already shows up in your daily routine, often without you realising it
- **Tools & Skills** - a practical overview of the tools used to work with and communicate data

Data Visuals

Different types of data visuals

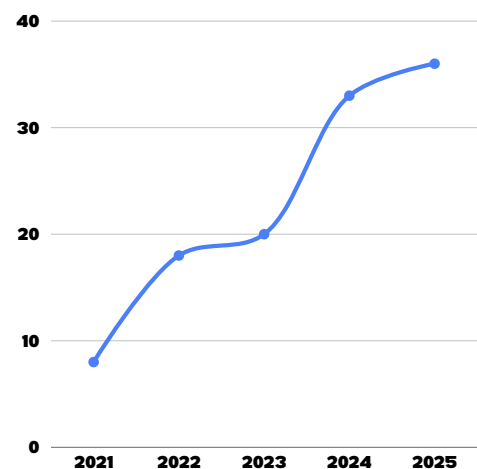
Data visualisation is the practice of representing information graphically – turning numbers and figures into charts and diagrams that make patterns, trends and relationships easier to see and understand. Being able to read and interpret these visuals makes it easier to make informed decisions, spot misleading information and engage confidently with the world around you. Knowing which visual to use, and how to read it, is a foundational skill for anyone navigating modern life.

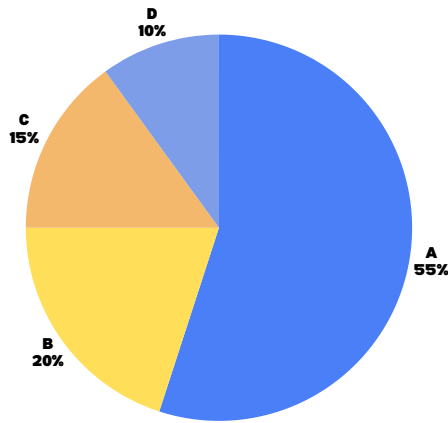
Below are the most common chart types you will encounter.



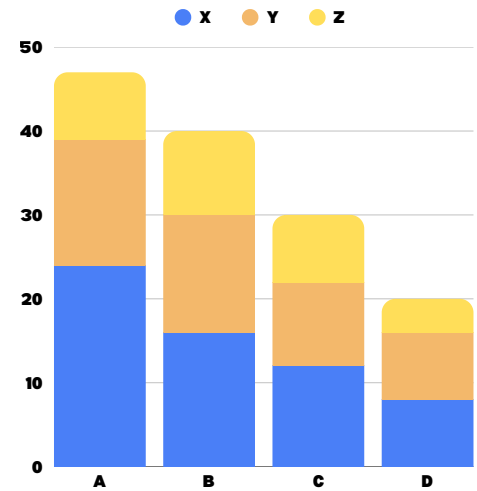
Bar Chart uses rectangular bars to compare values across different categories. The taller the bar, the greater the value.

Line Graph uses connected points to show how a value changes over time. Ideal for spotting trends and patterns.

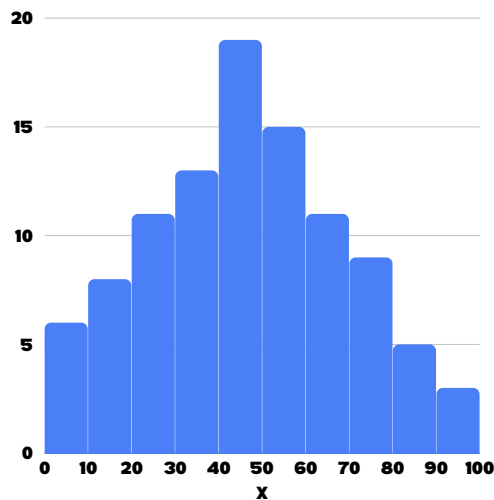




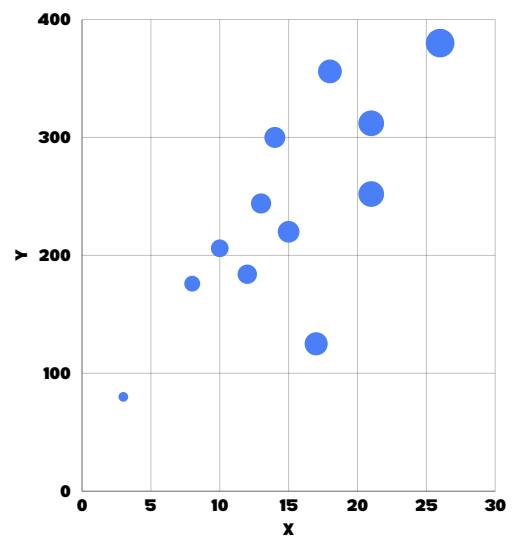
Pie Chart uses slices to show how a whole is divided into parts. Best used when comparing proportions.



Stacked Bar Chart uses layered bars to show how different parts contribute to a whole across multiple categories.



Histogram looks similar to a bar chart but shows how frequently values fall within a range. Used to display distribution.



Scatter Plot plots individual data points on two axes to reveal whether a relationship exists between two variables.

A simple guide to reading graphs/charts

Title

tells you what the chart is about. Always read this first as it frames everything you are about to see.

Axes

the horizontal axis (x) typically shows categories or time, while the vertical axis (y) shows values or quantities. Check what each axis represents before drawing any conclusions.

Scale

look at the numbers on the axes. A scale that does not start at zero can make differences appear larger than they actually are.

Legend

explains what colours, lines or patterns represent. Essential when a chart contains more than one variable.

Labels

identify specific data points, categories or units of measurement. They tell you exactly what you are looking at.

Source

indicates where the data came from. Always check this to assess the credibility and context of the information being presented.

Real Applications

Data is Already Part of Your Life!



Data is not something that lives in laboratories or corporate boardrooms. It is woven into the decisions you make every day - whether you notice it or not. Every time you check the weather before leaving the house, compare prices before a purchase, or glance at your step count before bed, you are engaging with data. The question is not whether data affects your life. It is whether you are reading it well enough to benefit from it.

1

The average smartphone user generates over 1GB of personal data every day through apps, location services, and browsing habits.

2

Supermarkets, streaming platforms, and social media feeds are all optimised using data visualisations that influence what you see, buy, and click.

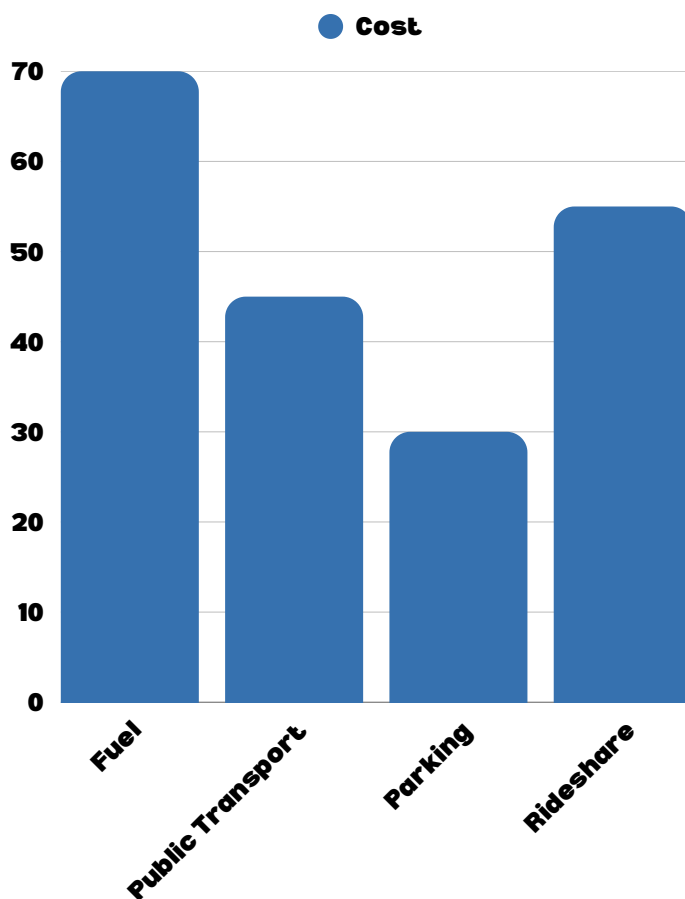
3

People who regularly track and visualise their personal data – spending, health, habits – consistently make better financial and lifestyle decisions than those who don't.

Getting Around is Costing You More Than You Think!

You know transport costs you something. You probably don't know how much.

Most people mentally account for their biggest transport expense – a monthly bus pass or fuel – and stop there. Parking, rideshare trips, and the occasional taxi get filed under "minor" and forgotten. But plotted together as a monthly bar chart, transport routinely emerges as one of the top three household expenses for young adults.



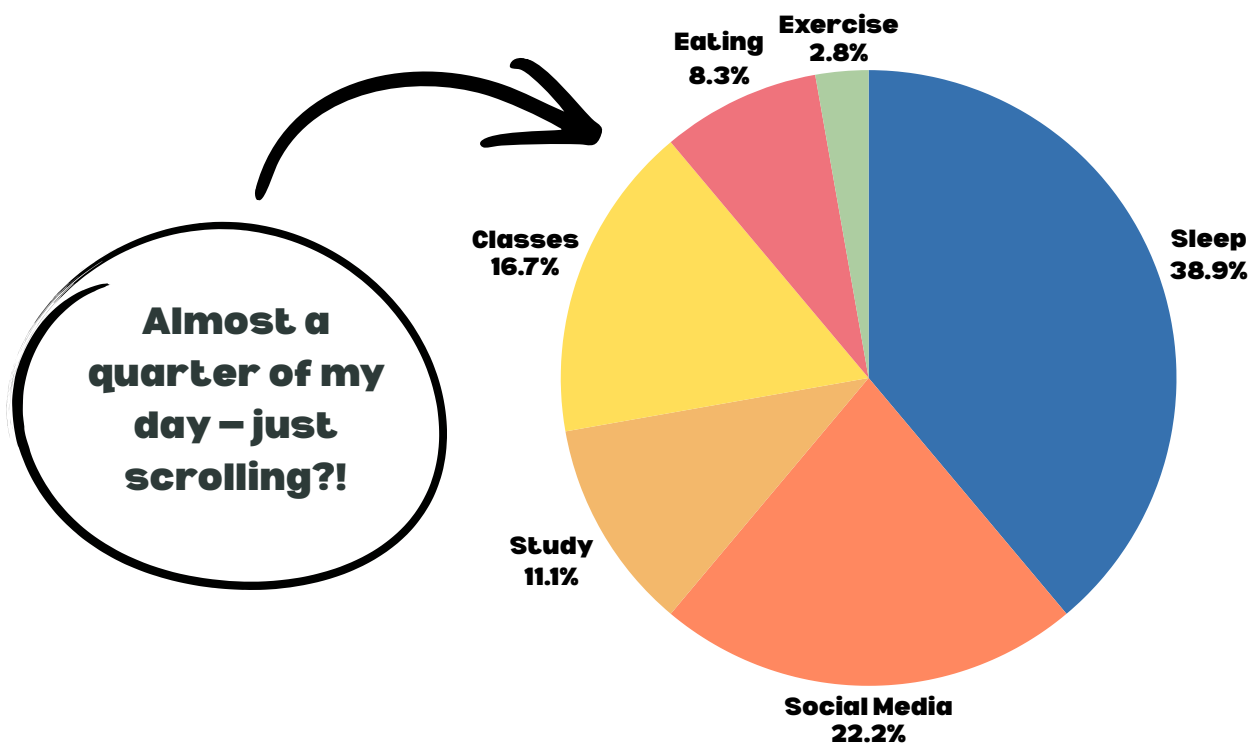
**Wait... I spend
more on Uber
than my bus
pass?**

Once you can see where your money is going, cutting back becomes obvious. Combine errands into one trip, swap a couple of rideshares for public transport, and watch that bar shrink!

Where Does Your Time Actually Go?

You know you use your phone a lot. You probably don't realise just how much.

Most students think of social media as something they do in between things – a quick scroll while waiting for a lecture, a few minutes before bed. But when you track your daily screen time and plot it as a pie chart, the picture looks very different. For most young adults, a single app accounts for more daily time than study, exercise, and cooking combined.

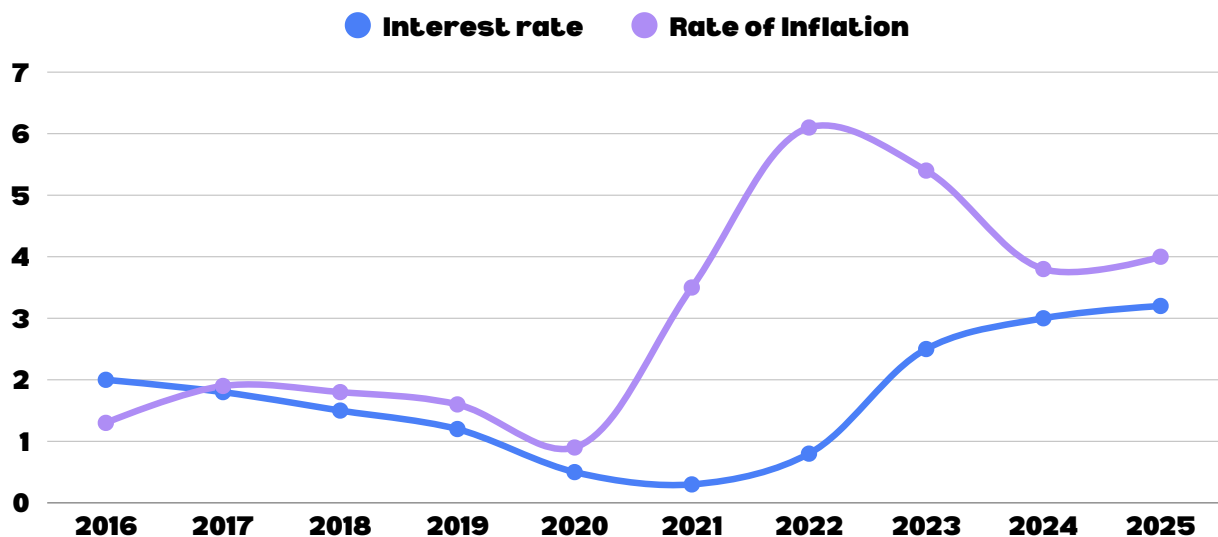


Once you can see how your time is actually split, small changes add up fast. Even cutting your screen time by 30 minutes a day gives you over three extra hours of study time each week!

Where Does Your Time Actually Go?

You have money in your savings account. The number is going up. So you're getting richer, right?

Not necessarily. Inflation is simply the rising cost of everyday things over time. When inflation rises faster than your savings account interest rate, your money buys less than it did last year – even if the number in your account looks bigger. Doing nothing with your money is not neutral. It is a slow loss.



**The lines crossed in 2017
and my money has been losing
ever since!**

You can avoid this by simply switching to a high interest savings account and staying on top of current inflation rates – because what keeps pace today might not keep pace next year.

Tools and Skills

It's simpler than you think!

In a world where data is everywhere, being able to read and create visualisations is quickly becoming as essential as knowing how to write an email. The good news is you do not need a degree in statistics or design to get started – you just need to know where to begin.

Start with your phone – your apps are already generating data visualisations for you every day, but understanding what they mean puts you in control rather than just along for the ride.

Use what you already have – Google Sheets and Excel can turn any table of numbers into a chart in under a minute; once you can build it yourself you stop relying on others

Copy before you create – find a chart you like and recreate it with your own data; the best way to develop your own eye for what works is by doing it yourself

Some Resources Curated Just for You!

Everything below is completely free and accessible right now – no subscriptions, no hidden costs. Whether you want to understand your spending, track your habits, or simply make sense of the data you encounter every day, these tools and resources will help you get started.

Google Sheets (sheets.google.com) – free with any Google account; turn your own spending, grades, or habits into a chart in minutes

Google Trends (trends.google.com) – see how search interest in any topic changes over time; great for spotting patterns in what people care about

Canva (canva.com) – free templates for charts and infographics; great for making your data look polished and shareable

Datawrapper (datawrapper.de) – build clean, professional charts with no coding required; perfect for visualising anything from your monthly budget to local news data

Pocketbook (pocketbook.com.au) – free Australian budgeting app that automatically turns your spending into visual breakdowns you can actually understand
Websites

Our World in Data (ourworldindata.org) – explore thousands of real world visualisations on health, money, climate, and more

Flowing Data (flowingdata.com) – approachable tutorials using everyday topics like food, fitness, and daily routines

ASIC's MoneySmart (moneysmart.gov.au) – free Australian government site with budgeting calculators and visual financial tools



Conclusion



Data has always been part of your everyday life. From understanding where your money goes, to recognising when a chart is trying to mislead you, the skills in this handbook are not abstract – they are immediately useful.

The world is generating more data than ever before, and the people who can make sense of it visually will always have an advantage. Not because they are smarter, but because they are paying attention in a way others are not.

You do not need to become a data scientist. You do not need to learn to code. You just need to start looking at the information already around you with a little more curiosity.



Pick one tool from the resources page and spend thirty minutes with it today



Next time you see a chart in the news or on social media, stop and ask who made it and why



Track something in your own life for two weeks – spending, study hours, screen time – and see what the data tells you

**The charts in this handbook showed you what is possible.
The next step is yours!**

Data visualisations are everywhere -in the news, at work, on social media and in everyday life. But do you know how to read them?

This guide breaks down the most common charts and graphs in a simple, accessible way.

Inside you will find:

- ✓ Clear explanations of common chart types
- ✓ Real life examples you will recognise
- ✓ Free online tools to help you create and explore your own charts

Whether you are a student, professional or just curious - this guide will give you the confidence to understand the data world around you.



Developed exclusively for research at the University of Tasmania

For any questions about the handbook content or resources mentioned, please feel free to get in touch:

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